PRACTICAL INSTRUMENTATION REFERENCE

OpenTelemetry GenAI Attribute Checklist

Short edition

SOURCES semantic-conventions-genai `c88d504ab3d9`
semantic-conventions `d0472f4ae331`
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REQUIRED Set for every matching signal. CONDITIONALLY REQUIRED Set when the source condition applies. RECOMMENDED Set when available or applicable.

LLM call (inference) span

Inference

SOURCE ↗

SPAN NAME SHOULD be {gen_ai.operation.name} {gen_ai.request.model}. Semantic conventions for individual GenAI systems and frameworks MAY specify different span name format and MUST follow the overall guidelines for span names.

SPAN KIND SHOULD be CLIENT and MAY be set to INTERNAL on spans representing call to models running in the same process. It's RECOMMENDED to use CLIENT kind when the GenAI system being instrumented usually runs in a different process than its client or when the GenAI call happens over instrumented protocol such as HTTP.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION
`gen_ai.operation.name`	string	REQUIRED	The name of the operation being performed.
`gen_ai.provider.name`	string	REQUIRED	The Generative AI provider as identified by the client or server…
`error.type`	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.
`gen_ai.conversation.id`	string	CONDITIONALLY REQUIRED	The unique identifier for a conversation (session, thread), used to store…
`gen_ai.output.type`	string	CONDITIONALLY REQUIRED	Represents the content type requested by the client.
`gen_ai.prompt.name`	string	CONDITIONALLY REQUIRED when a named prompt template is used	The name of the prompt that uniquely identifies it.
`gen_ai.prompt.version`	string	CONDITIONALLY REQUIRED	The version of the prompt template used.
`gen_ai.request.choice.count`	int	CONDITIONALLY REQUIRED If available, in the request, and !=1	The target number of candidate completions to return.
`gen_ai.request.model`	string	CONDITIONALLY REQUIRED If available	The name of the GenAI model a request is being made to.
`gen_ai.request.seed`	int	CONDITIONALLY REQUIRED If applicable and if the request includes a seed	Requests with same seed value more likely to return same result.
`gen_ai.request.stream`	boolean	CONDITIONALLY REQUIRED	Indicates whether the GenAI request was made in streaming mode.
`gen_ai.request.top_k`	int	CONDITIONALLY REQUIRED If applicable	The top-K sampling setting for the GenAI request: restricts token…
`server.port`	int	CONDITIONALLY REQUIRED If server.address is set	GenAI server port.
`gen_ai.conversation.compacted`	boolean	RECOMMENDED when available	Indicates whether the effective conversation context used for this…
`gen_ai.request.frequency_penalty`	double	RECOMMENDED	The frequency penalty setting for the GenAI request.
`gen_ai.request.max_tokens`	int	RECOMMENDED	The maximum number of tokens the model generates for a request.
`gen_ai.request.presence_penalty`	double	RECOMMENDED	The presence penalty setting for the GenAI request.
`gen_ai.request.previous_response.id`	string	RECOMMENDED	The unique identifier of a previous response or interaction used to provide…
`gen_ai.request.reasoning.level`	string	RECOMMENDED When applicable	The reasoning or thinking effort level requested for a GenAI model.
`gen_ai.request.stop_sequences`	string[]	RECOMMENDED	List of sequences that the model will use to stop generating further tokens.
`gen_ai.request.temperature`	double	RECOMMENDED	The temperature setting for the GenAI request.
`gen_ai.request.top_p`	double	RECOMMENDED	The top_p sampling setting for the GenAI request.
`gen_ai.response.finish_reasons`	string[]	RECOMMENDED	Array of reasons the model stopped generating tokens, corresponding to each…
`gen_ai.response.id`	string	RECOMMENDED	The unique identifier for the completion.
`gen_ai.response.model`	string	RECOMMENDED	The name of the model that generated the response.
`gen_ai.response.time_to_first_chunk`	double	RECOMMENDED If the request was a streaming request	Time to first chunk in a streaming response, measured from request…
`gen_ai.usage.audio.cache_read.input_tokens`	int	RECOMMENDED When applicable	The number of audio input tokens served from a provider-managed cache.
`gen_ai.usage.audio.input_tokens`	int	RECOMMENDED When applicable	The number of audio input tokens.
`gen_ai.usage.audio.output_tokens`	int	RECOMMENDED When applicable	The number of audio output tokens.
`gen_ai.usage.cache_read.input_tokens`	int	RECOMMENDED When applicable	The number of input tokens served from a provider-managed cache.
`gen_ai.usage.cache_write.input_tokens`	int	RECOMMENDED When applicable	The number of input tokens written to a provider-managed cache.
`gen_ai.usage.image.cache_read.input_tokens`	int	RECOMMENDED When applicable	The number of image input tokens served from a provider-managed cache.
`gen_ai.usage.image.input_tokens`	int	RECOMMENDED When applicable	The number of image input tokens.
`gen_ai.usage.image.output_tokens`	int	RECOMMENDED When applicable	The number of image output tokens.
`gen_ai.usage.input_tokens`	int	RECOMMENDED	The number of tokens used in the GenAI input (prompt).
`gen_ai.usage.output_tokens`	int	RECOMMENDED	The number of tokens used in the GenAI response (completion).
`gen_ai.usage.reasoning.output_tokens`	int	RECOMMENDED When applicable	The number of output tokens used for reasoning (e.g. chain-of-thought…
`gen_ai.usage.text.cache_read.input_tokens`	int	RECOMMENDED When applicable	The number of text input tokens served from a provider-managed cache.
`gen_ai.usage.text.input_tokens`	int	RECOMMENDED When applicable	The number of text input tokens.
`gen_ai.usage.text.output_tokens`	int	RECOMMENDED When applicable	The number of text output tokens.
`server.address`	string	RECOMMENDED	GenAI server address.

Tool execution span

Execute tool

SOURCE ↗

SPAN NAME SHOULD be execute_tool {gen_ai.tool.name}.

SPAN KIND SHOULD be INTERNAL.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION
gen_ai.operation.name	string	REQUIRED	The name of the operation being performed.
gen_ai.tool.name	string	REQUIRED	Name of the tool utilized by the agent.
error.type	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.
gen_ai.agent.name	string	CONDITIONALLY REQUIRED When applicable	The human-readable name of the agent executing the tool.
gen_ai.conversation.id	string	CONDITIONALLY REQUIRED If available	The unique identifier for a conversation (session, thread), used to store…
gen_ai.tool.call.id	string	RECOMMENDED If available	The tool call identifier.
gen_ai.tool.description	string	RECOMMENDED If available	The tool description.
gen_ai.tool.type	string	RECOMMENDED If available	Type of the tool utilized by the agent

Agent spans

Create agent span

SOURCE ↗

SPAN NAME SHOULD be create_agent {gen_ai.agent.name}. Semantic conventions for individual GenAI systems and frameworks MAY specify different span name format.

SPAN KIND SHOULD be CLIENT.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION
gen_ai.operation.name	string	REQUIRED	The name of the operation being performed.
gen_ai.provider.name	string	REQUIRED	The Generative AI provider as identified by the client or server…
error.type	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.
gen_ai.agent.description	string	CONDITIONALLY REQUIRED If provided by the application	The free-form description of the GenAI agent created during this operation.
gen_ai.agent.id	string	CONDITIONALLY REQUIRED If applicable	The stable unique identifier of the GenAI agent created during this operation.
gen_ai.agent.name	string	CONDITIONALLY REQUIRED If provided by the application	The human-readable name of the GenAI agent created during this operation.
gen_ai.agent.version	string	CONDITIONALLY REQUIRED If provided by the application	The version of the GenAI agent created during this operation.
gen_ai.request.model	string	CONDITIONALLY REQUIRED If available	The name of the GenAI model a request is being made to.
server.port	int	CONDITIONALLY REQUIRED If server.address is set	GenAI server port.
server.address	string	RECOMMENDED	GenAI server address.

Invoke agent client span

SOURCE ↗

SPAN NAME SHOULD be invoke_agent {gen_ai.agent.name} if gen_ai.agent.name is readily available. When gen_ai.agent.name is not available, it SHOULD be invoke_agent. Semantic conventions for individual GenAI systems and frameworks MAY specify different span name format.

SPAN KIND SHOULD be CLIENT.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION
gen_ai.operation.name	string	REQUIRED	The name of the operation being performed.
gen_ai.provider.name	string	REQUIRED	The Generative AI provider as identified by the client or server…
error.type	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.
gen_ai.agent.description	string	CONDITIONALLY REQUIRED When available	The free-form description of the invoked GenAI agent.
gen_ai.agent.id	string	CONDITIONALLY REQUIRED If applicable	The stable unique identifier of the invoked GenAI agent.
gen_ai.agent.name	string	CONDITIONALLY REQUIRED When available	The human-readable name of the invoked GenAI agent.
gen_ai.agent.version	string	CONDITIONALLY REQUIRED When available	The version of the invoked GenAI agent.
gen_ai.conversation.id	string	CONDITIONALLY REQUIRED	The unique identifier for a conversation (session, thread), used to store…
gen_ai.data_source.id	string	CONDITIONALLY REQUIRED If applicable	The data source identifier.
gen_ai.output.type	string	CONDITIONALLY REQUIRED	Represents the content type requested by the client.
gen_ai.request.choice.count	int	CONDITIONALLY REQUIRED If available, in the request, and !=1	The target number of candidate completions to return.
gen_ai.request.seed	int	CONDITIONALLY REQUIRED If applicable and if the request includes a seed	Requests with same seed value more likely to return same result.
server.port	int	CONDITIONALLY REQUIRED If server.address is set	GenAI server port.
gen_ai.request.frequency_penalty	double	RECOMMENDED	The frequency penalty setting for the GenAI request.
gen_ai.request.max_tokens	int	RECOMMENDED	The maximum number of tokens the model generates for a request.
gen_ai.request.model	string	RECOMMENDED If applicable	The name of the GenAI model configured for the agent.
gen_ai.request.presence_penalty	double	RECOMMENDED	The presence penalty setting for the GenAI request.
gen_ai.request.previous_response.id	string	RECOMMENDED	The unique identifier of a previous response or interaction used to provide…
gen_ai.request.stop_sequences	string[]	RECOMMENDED	List of sequences that the model will use to stop generating further tokens.
gen_ai.request.temperature	double	RECOMMENDED	The temperature setting for the GenAI request.
gen_ai.request.top_p	double	RECOMMENDED	The top_p sampling setting for the GenAI request.
gen_ai.response.finish_reasons	string[]	RECOMMENDED	Array of reasons the model stopped generating tokens, corresponding to each…
gen_ai.usage.audio.cache_read.input_tokens	int	RECOMMENDED When applicable	The number of audio input tokens served from a provider-managed cache.
gen_ai.usage.audio.input_tokens	int	RECOMMENDED When applicable	The number of audio input tokens.
gen_ai.usage.audio.output_tokens	int	RECOMMENDED When applicable	The number of audio output tokens.
gen_ai.usage.cache_read.input_tokens	int	RECOMMENDED When applicable	The number of input tokens served from a provider-managed cache.
gen_ai.usage.cache_write.input_tokens	int	RECOMMENDED When applicable	The number of input tokens written to a provider-managed cache.
gen_ai.usage.image.cache_read.input_tokens	int	RECOMMENDED When applicable	The number of image input tokens served from a provider-managed cache.
gen_ai.usage.image.input_tokens	int	RECOMMENDED When applicable	The number of image input tokens.
gen_ai.usage.image.output_tokens	int	RECOMMENDED When applicable	The number of image output tokens.
gen_ai.usage.input_tokens	int	RECOMMENDED	The number of tokens used in the GenAI input (prompt).
gen_ai.usage.output_tokens	int	RECOMMENDED	The number of tokens used in the GenAI response (completion).
gen_ai.usage.text.cache_read.input_tokens	int	RECOMMENDED When applicable	The number of text input tokens served from a provider-managed cache.
gen_ai.usage.text.input_tokens	int	RECOMMENDED When applicable	The number of text input tokens.
gen_ai.usage.text.output_tokens	int	RECOMMENDED When applicable	The number of text output tokens.
server.address	string	RECOMMENDED	GenAI server address.

EXPLICITLY ENABLED CONTENT

Capturing content (opt-in)

Source caution. Model instructions, user messages, and model outputs are considered sensitive and are often large in size. OpenTelemetry instrumentations SHOULD NOT capture them by default, but SHOULD provide an option for users to opt in.

SOURCE ↗

	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SOURCE DESCRIPTION
☐	`gen_ai.input.messages` DEVELOPMENT	any	OPT-IN	The chat history provided to the model as an input.
☐	`gen_ai.output.messages` DEVELOPMENT	any	OPT-IN	Messages returned by the model where each message represents a specific model response (choice, candidate).
☐	`gen_ai.system_instructions` DEVELOPMENT	any	OPT-IN	The system message or instructions provided to the GenAI model separately from the chat history.
☐	`gen_ai.tool.definitions` DEVELOPMENT	any	OPT-IN	The list of tool definitions available to the GenAI agent or model.
☐	`gen_ai.tool.call.arguments` DEVELOPMENT	any	OPT-IN	Parameters passed to the tool call.
☐	`gen_ai.tool.call.result` DEVELOPMENT	any	OPT-IN	The result returned by the tool call (if any and if execution was successful).

Input/output messages.

Messages MUST be provided in the order they were sent to the model. Each output message represents a single output choice/candidate generated by the model.

Structured values.

If structured attributes are not yet supported on spans in a given language, the corresponding value SHOULD be serialized to JSON string on spans and recorded in structured form on events.

Tool content.

For tool definitions, tool call arguments, and tool call results, the source states: "This attribute may contain sensitive information."

Metrics

gen_ai.client.token.usage

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	{token}	DEVELOPMENT	Number of input and output tokens used.

KEY ATTRIBUTE	REQUIREMENT
gen_ai.operation.name	REQUIRED
gen_ai.provider.name	REQUIRED
gen_ai.token.type	REQUIRED
gen_ai.request.model	CONDITIONALLY REQUIRED If available
server.port	CONDITIONALLY REQUIRED If server.address is set
gen_ai.response.model	RECOMMENDED
server.address	RECOMMENDED

gen_ai.client.operation.duration

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	GenAI operation duration.

KEY ATTRIBUTE	REQUIREMENT
gen_ai.operation.name	REQUIRED
error.type	CONDITIONALLY REQUIRED If the operation ended in an error
gen_ai.provider.name	CONDITIONALLY REQUIRED
gen_ai.request.model	CONDITIONALLY REQUIRED If available
server.port	CONDITIONALLY REQUIRED If server.address is set
gen_ai.response.model	RECOMMENDED
server.address	RECOMMENDED

Deprecated - do not use

OLD REGISTRY SOURCE ↗

NAME	REPLACEMENT	SOURCE NOTE
gen_ai.completion	Deprecated, use Event API to report completions contents.	Removed, no replacement at this time.
gen_ai.prompt	Deprecated, use Event API to report prompt contents.	Removed, no replacement at this time.
gen_ai.system	Deprecated, use gen_ai.provider.name instead.	Replaced by gen_ai.provider.name, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.usage.completion_tokens	Deprecated, use gen_ai.usage.output_tokens instead.	Replaced by gen_ai.usage.output_tokens, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.usage.prompt_tokens	Deprecated, use gen_ai.usage.input_tokens instead.	Replaced by gen_ai.usage.input_tokens, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.openai.request.response_format	Deprecated, use gen_ai.output.type.	Replaced by gen_ai.output.type, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.openai.request.seed	Deprecated, use gen_ai.request.seed.	Replaced by gen_ai.request.seed, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.openai.request.service_tier	Deprecated, use openai.request.service_tier.	Replaced by openai.request.service_tier, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.openai.response.service_tier	Deprecated, use openai.response.service_tier.	Replaced by openai.response.service_tier, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.openai.response.system_fingerprint	Deprecated, use openai.response.system_fingerprint.	Replaced by openai.response.system_fingerprint, which has moved to the OpenTelemetry GenAI semantic conventions repository.

Full reference: OpenTelemetry GenAI Attribute Checklist.pdf (47+ pages)